Classifier Evaluation

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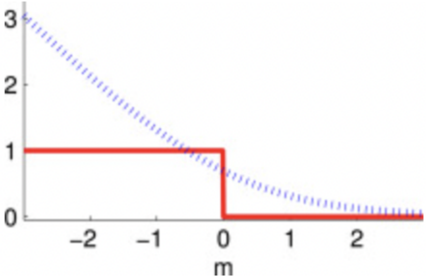


Evaluation of a Classifier/Model/System

- How **good** is it?
- Levels of **goodness**
  - **Absolute goodness:** When we run our trained model in the "wild" it does what we expect it to do
  - no way of knowing *before* we deploy the model and when we do know, too late!
- **Relative goodness:** We have a small representative sample of **test data**
  - We compare the output produced by our classifier on this test dataset (*gold standard*) and measure how well it resembles the labels in the dataset

Gold Standard

- A dataset that we use for evaluation purpose. Also known as **test data**.
- Each test instance in the test data has its correct label annotated.
- Numerous measures exist (as we will shortly see) to **compare** the **predicted** labels by the trained classifier and actual (**target**) labels in the test dataset.
- **Never train on test data!!!**

Confusion Matrix (error matrix)

		Actual	
		Actual YES(+)	**Actual NO(-)**
Predicted	**Predicted YES(+)**	**True Positives (TP)**	**False Positives (FP)**
	Predicted NO(-)	**False Negatives (FN)**	**True Negatives (TN)**

The matrix makes it easy to see if the system is **confusing** two classes.

Definitions

- **True Positive**

  We predicted as positive and it is indeed positive
- **True Negative**

  We predicted as negative and it is indeed negative
- **False Positive**

  We predicted as positive but it turns out to be negative
- **False Negative**

  We predicted as negative but it turns out to be positive

		Actual	
		Actual YES(+)	**Actual** NO(-)
Predicted	**Predicted** YES(+)	**True Positives** (TP)	**False Positives** (FP)
	Predicted NO(-)	**False Negatives** (FN)	**True Negatives** (TN)

Example

- We have a set of images some of which contain a car
- We want to detect photos with a car
- Positive class (+): all photos with a car
- Negative class (-): all other photos

Example: Detecting Cancer

- Let assume we trained a classifier to detect cancer based on some features.
- Predicting YES means we predict that the patient has cancer.
- We predicted the patient as having cancer but further tests revealed that the patient does not have cancer
  - False Positive
- We predicted the patient as not having cancer (so no further tests were done) but the patient died with cancer!
  - False Negative
- The moral of the story
  - FP and FN have very different importance in real-world data mining tasks.

Evaluation Measures

- Accuracy
- Precision
- Recall
- F-score
- Many more (see e.g. Confusion matrix)

		Actual	
		Actual YES(+)	**Actual** NO(-)
Predicted	**Predicted** YES(+)	**True Positives** (TP)	**False Positives** (FP)
	Predicted NO(-)	**False Negatives** (FN)	**True Negatives** (TN)

Evaluation Measures: Accuracy

Answers to

what proportion of all objects were **correctly classified**?

$$\text{Accuracy} = \frac{\text{TP} + \text{TN}}{\text{TP} + \text{TN} + \text{FP} + \text{FN}}$$

Actual

Predicted

	Actual YES(+)	**Actual** NO(-)
Predicted YES(+)	**True Positives** (TP)	**False Positives** (FP)
Predicted NO(-)	**False Negatives** (FN)	**True Negatives** (TN)

Evaluation Measures: Precision

Answers to

what proportion of **predicted Positives** is truly Positive?

$$\text{Precision} = \frac{\text{TP}}{\text{TP} + \text{FP}}$$

Actual

Predicted	**Actual** YES(+)	**Actual** NO(-)
Predicted YES(+)	**True Positives** (TP)	**False Positives** (FP)
Predicted NO(-)	**False Negatives** (FN)	**True Negatives** (TN)

Evaluation Measures: Recall

Answers to

what proportion of **actual Positives** is correctly classified?

$$\text{Recall} = \frac{\text{TP}}{\text{TP} + \text{FN}}$$

		Actual	
		Actual YES(+)	**Actual** NO(-)
Predicted	**Predicted** YES(+)	**True Positives** (TP)	**False Positives** (FP)
	Predicted NO(-)	**False Negatives** (FN)	**True Negatives** (TN)

What is more important: Precision or Recall?

Application 1 (cancer detection)

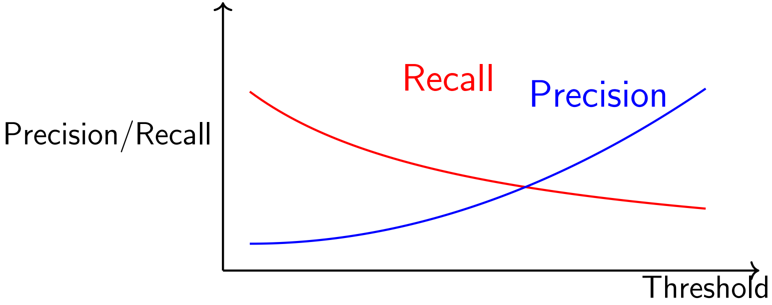
- Positive: has cancer
- Negative: healthy
- We want to detect as many patients with cancer as possible
- We want to have **high recall**

Application 2 (product recommendation)

- Positive: relevant products
- Negative: non-relevant products
- We want to make sure that almost all recommended products are relevant to the user
- We want to have **high precision**

There is a **trade-off** between precision and recall: improving precision often results in lowering recall and vice versa

Precision–Recall Trade-off



By simply varying the **threshold** of our cancer detector we can get a **high precision** OR **low recall** system. There is a **trade-off**.

Evaluation measures: F-score

$$\text{F-score} = \frac{2 \times \text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}}$$

F-score is the **harmonic mean** between precision and recall

$$\text{F-score} = \frac{2}{\frac{1}{\text{Precision}} + \frac{1}{\text{Recall}}}$$

Actual

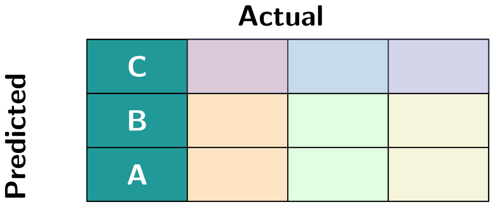
Predicted	**Actual** YES(+)	**Actual** NO(-)
Predicted YES(+)	**True Positives** (TP)	**False Positives** (FP)
Predicted NO(-)	**False Negatives** (FN)	**True Negatives** (TN)

- F-score is in between precision and recall
- F-score gives a larger weight to lower numbers

Evaluation Measures for Multiple Classes

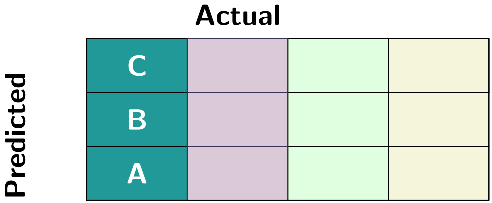
Precision for a class A:

$$\frac{\text{no. objects correctly classified } A}{\text{no. objects classified } A}$$



Recall for class A:

$$\frac{\text{no. objects correctly classified } A}{\text{no. objects that belong to class } A}$$



Evaluation Measures for Multiple Classes

F-score for class A

$$\text{F-score}_A = \frac{2 \times \text{Precision}_A \times \text{Recall}_A}{\text{Precision}_A + \text{Recall}_A}$$

Macro F-score

$$\text{Macro F-score} = \frac{1}{C} \sum_{i=1}^{C} \text{F-score}_i$$

where C is the number of classes and F-score_i is the F-score for class i.